

PROJECT BUTTON HR–V0 DXL BRANCH–PROTECTION EVALUATION

Forward current limiting plus short–pulse regenerative clamp candidate; reverse–current proof remains open.

01 Branch 1 TPS259461L evaluation circuit

File: 01_branch_1.kicad_sch

02 Branch 2 TPS259461L evaluation circuit

File: 02_branch_2.kicad_sch

03 Branch 3 TPS259461L evaluation circuit

File: 03_branch_3.kicad_sch

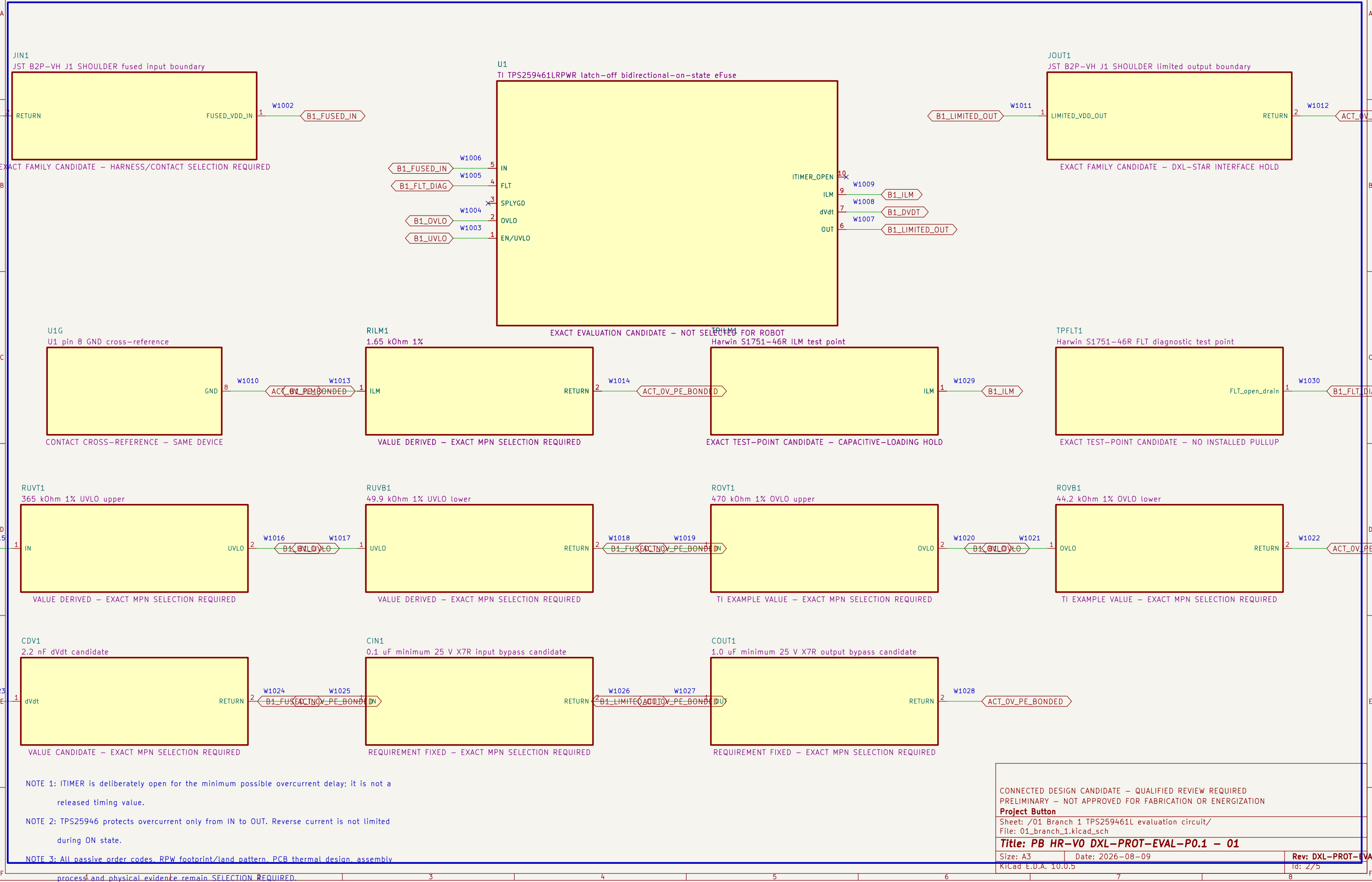
04 Source and regenerative–clamp evaluation boundary

File: 04_regeneration_clamp.kicad_sch

V2.1 PRESERVED; V3 IS A CONNECTED CANDIDATE		
PRELIMINARY – NOT APPROVED FOR FABRICATION OR ENERGIZATION		
Project Button		
Sheet: /		
File: hr–v0–dxl–protection–eval.kicad_sch		
Title: PROJECT BUTTON HR–V0 DXL BRANCH–PROTECTION EVALUATION		
Size: A3	Date: 2026–08–09	Rev: DXL–PROT–EVAL–P0.1
KiCad E.D.A. 10.0.5	Id: 1/5	

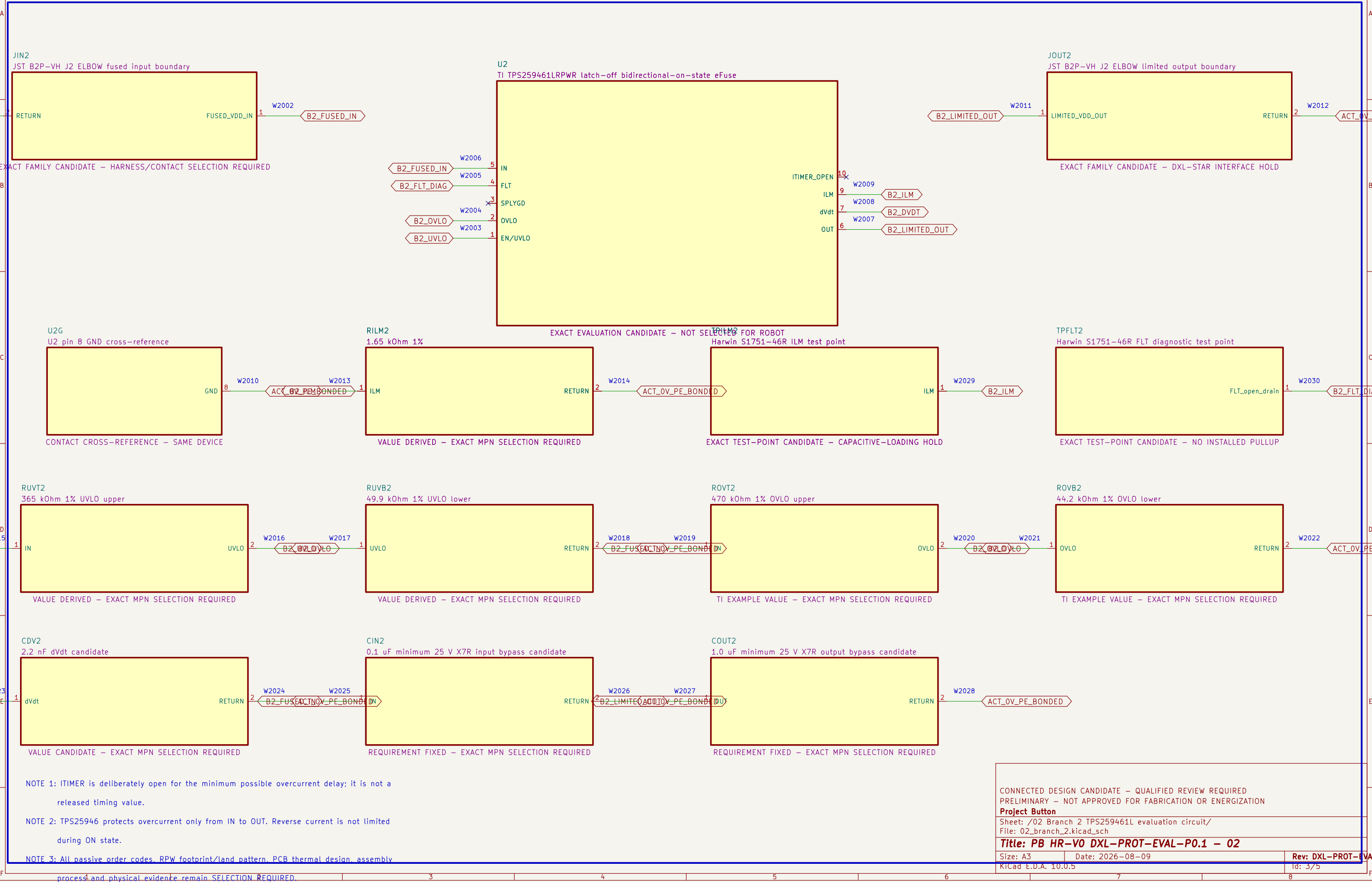
01 Branch 1 TPS259461L evaluation circuit

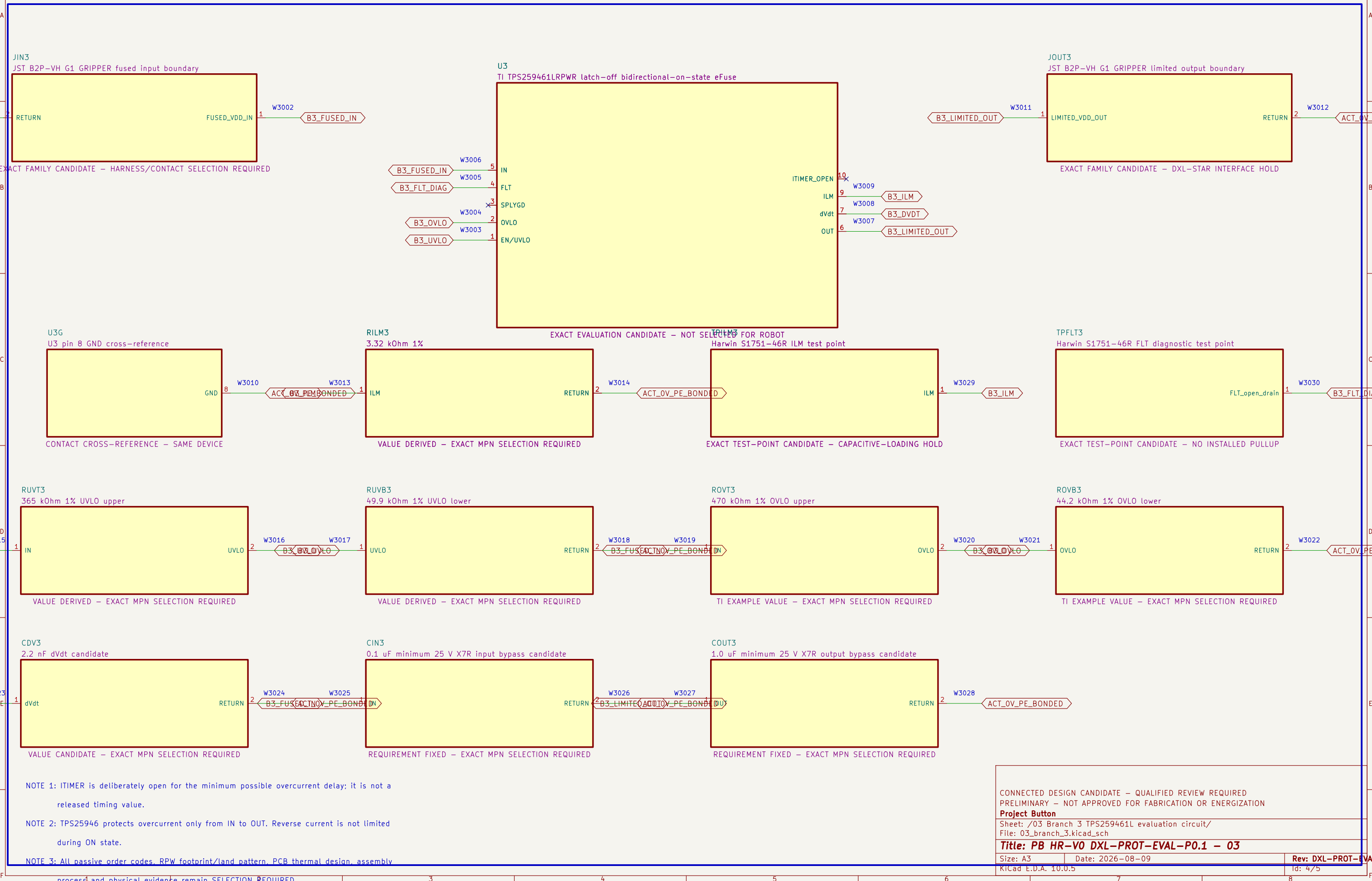
Exact eFuse pinout with derived dividers, current setting, bypassing and diagnostic test points.



02 Branch 2 TPS259461L evaluation circuit

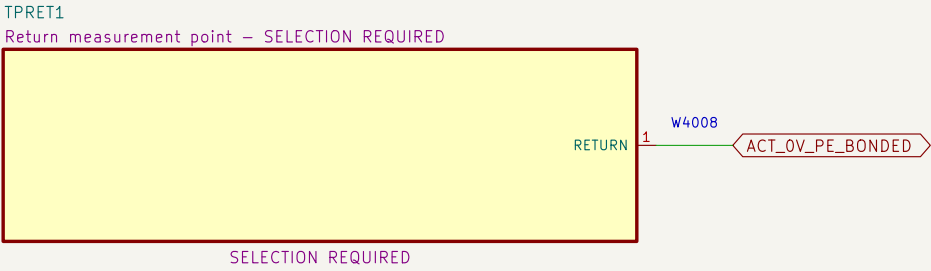
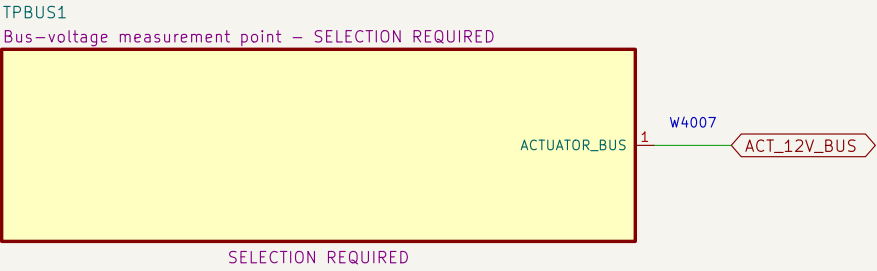
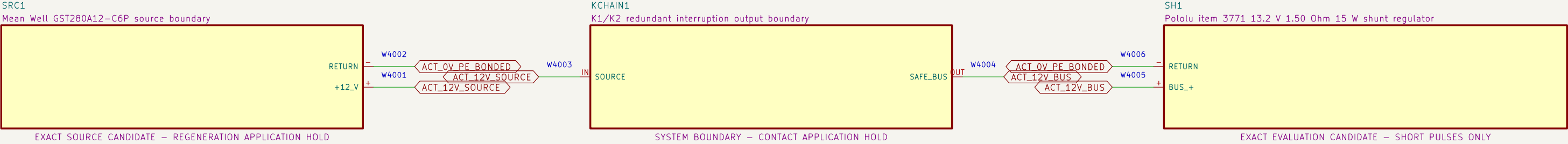
Exact eFuse pinout with derived dividers, current setting, bypassing and diagnostic test points.





04 Source and regenerative–clamp evaluation boundary

The short–pulse shunt candidate is across the post–contactor bus; source sink behavior remains unproved.



NOTE 1: Pololu item 3771 is a short–pulse evaluation candidate, not a continuous dump load or safety function.

NOTE 2: The 12 V source may reach 12.6 V by tolerance; the shunt setpoint may be 12.804 V at –3%. Nuisance margin requires received measurement.

NOTE 3: Contact opening, eFuse UVLO/OVLO action, stored energy and three–axis simultaneous

CONNECTED DESIGN CANDIDATE – QUALIFIED REVIEW REQUIRED		
PRELIMINARY – NOT APPROVED FOR FABRICATION OR ENERGIZATION		
Project Button		
Sheet: /04 Source and regenerative–clamp evaluation boundary/ File: 04_regeneration_clamp.kicad_sch		
Title: PB HR–VO DXL–PROT–EVAL–P0.1 – 04		
Size: A3	Date: 2026–08–09	Rev: DXL–PROT–EVAL–P0.1
KiCad E.D.A. 10.0.5	Id: 5/5	